Orbital Speed Equation

Exam code: 0625 & 0972

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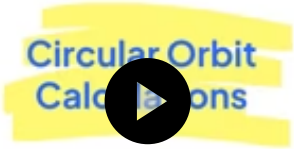

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Orbital speed equation

Help me understand

Extended tier only

- When planets orbit around the Sun, or a moon moves around a planet, they move in **circular orbits**
- In one complete orbit, a planet travels a distance equal to the circumference of a circle
  - This is equal to $2\pi r$, where r is the radius of the circular path
- The relationship between speed, distance and time is:

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{\text{circumference of orbit}}{\text{orbital period}}$$

- The average orbital speed of an object can be defined by the equation:

$$v = \frac{2\pi r}{T}$$

- Where:
  - v = orbital speed in metres per second (m/s)
  - r = average radius of the orbit in metres (m)
  - T = orbital period in seconds (s)
- This orbital period (or time period) is defined as:

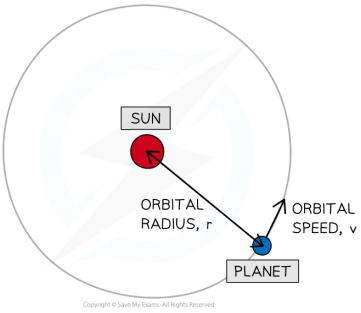
The time taken for an object to complete one orbit

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Orbital speed of a planet



Orbital radius and orbital speed of a planet moving around a Sun

Worked Example

The Hubble Space Telescope moves in a circular orbit. Its height above the Earth's surface is 560 km and the radius of the Earth is 6400 km. It completes one orbit in 96 minutes.

Calculate its orbital speed in m/s.

Answer:

Step 1: List the known quantities

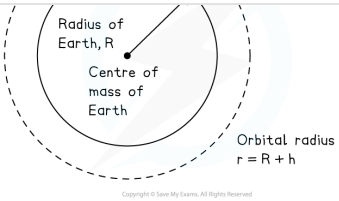
- Radius of the Earth, R = 6400 km
- Height of the telescope above the Earth's surface, h = 560 km
- Time period, T = 96 minutes

Step 2: Write the relevant equation

$$v = \frac{2\pi r}{T}$$

Step 3: Calculate the orbital radius, *r*

- The **orbital radius** is the distance from the **centre** of the Earth to the telescope



$$r = R + h$$

$$r = 6400 + 560 = 6960 \text{ km}$$

Step 4: Convert any units

- The time period needs to be in seconds

$$1 \text{ minute} = 60 \text{ seconds}$$

$$96 \text{ minutes} = 60 \times 96 = 5760 \text{ s}$$

- The radius needs to be in metres

$$1 \text{ km} = 1000 \text{ m}$$

$$6960 \text{ km} = 6\,960\,000 \text{ m}$$

Step 5: Substitute values into the orbital speed equation

$$v = \frac{2\pi \times 6\,960\,000}{5760} = 7592.18 = 7590 \text{ m/s}$$

Examiner Tips and Tricks

Remember to check that the orbital radius *r* given is the distance from the **centre** of the Sun (if a planet is orbiting a Sun) or the planet (if a moon is orbiting a planet) and not just from the surface. If the distance is a height above the surface you must add the radius of the body, to get the height above the centre of mass of the body.

This is because orbits are caused by the mass, which can be assumed to act at the centre, rather than the surface.

Don't forget to check your units and convert any if required!

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